

MACHILIPATNAM, IndiaWMO: **431850**Lat: **16.1829N**Lon: **81.1533E**Elev: **3**StdP: **101.29**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	20.2	15.2	10.8	26.7	16.4	11.7	26.0	5.5	26.2	4.9	26.3	1.0	0	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.5	40.6	26.7	38.2	26.9	36.6	27.1	29.4	34.8	29.0	34.0	28.6	33.4	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.2	24.3	32.3	27.7	23.7	31.9	27.4	23.2	31.5	97.0	35.3	94.8	34.2	92.7	33.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
6.2	5.2	4.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			16.9	44.2	2.2	1.9	15.3	45.6	14.0	46.7	12.8	47.8	11.3	49.2
			15.8	30.3	0.9	0.6	15.2	30.8	14.7	31.2	14.2	31.5	13.5	32.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	24.8	26.4	28.6	30.8	32.9	31.9	30.1	29.6	29.6	28.5	26.8	25.1
	DBStd	2.99	1.07	1.16	1.20	1.28	2.23	2.59	2.09	1.81	1.55	1.50	1.33	1.23
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6853	460	458	576	625	711	658	624	609	587	574	503	467
	CDD18.3	3811	202	224	318	375	453	408	366	351	337	316	253	208
	CDH23.3	45427	1339	1942	3629	5062	6693	5844	4761	4415	4242	3642	2377	1480
CDH26.7	21753	303	656	1568	2714	4228	3494	2382	2054	1929	1398	691	335	
Wind	WSAvg	1.7	1.3	1.4	1.7	2.3	2.3	2.2	1.9	1.7	1.4	1.3	1.6	1.5
Precipitation	PrecAvg	1087	5	12	8	11	37	115	182	171	178	210	119	21
	PrecMax	2092	49	95	135	179	233	330	450	417	543	764	546	227
	PrecMin	499	0	0	0	0	0	15	45	20	10	19	0	0
	PrecStd	305	10	25	25	27	50	77	90	88	110	156	133	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	33.7	35.3	37.2	44.4	42.4	38.2	36.8	36.0	34.9	32.8	31.3
		MCWB	23.6	24.2	26.2	27.8	25.3	27.0	25.9	26.5	26.8	26.5	25.1	23.7
	2%	DB	30.0	32.2	34.1	36.1	41.9	40.0	36.7	35.2	34.9	33.6	31.5	30.0
		MCWB	23.1	24.1	26.1	27.8	25.9	26.8	25.9	26.5	26.9	26.6	24.8	23.4
	5%	DB	29.1	31.2	33.3	35.3	39.1	38.1	35.3	34.1	33.8	32.7	30.7	29.3
		MCWB	22.7	24.0	25.9	27.6	26.9	26.8	26.2	26.7	27.0	26.5	24.7	22.9
	10%	DB	28.4	30.2	32.5	34.5	36.8	36.3	33.9	33.1	32.8	31.7	29.9	28.6
		MCWB	22.5	23.8	25.6	27.4	27.9	26.7	26.3	26.5	26.9	26.4	24.5	22.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	26.7	27.8	29.2	30.3	29.8	28.5	28.4	28.6	28.3	27.0	25.2
		MCDB	29.1	31.2	33.1	34.7	36.3	35.9	33.3	32.7	32.7	32.2	30.5	29.0
	2%	WB	24.3	25.6	27.2	28.7	29.6	29.1	28.0	27.9	28.0	27.7	26.4	24.7
		MCDB	28.0	30.2	32.3	33.9	35.1	34.6	32.5	32.0	32.1	31.6	29.7	28.3
	5%	WB	23.7	25.0	26.8	28.3	29.2	28.6	27.6	27.5	27.7	27.3	26.0	24.2
		MCDB	27.4	29.4	31.5	33.3	34.3	33.7	31.8	31.5	31.5	31.0	29.2	27.6
	10%	WB	23.2	24.4	26.3	27.9	28.8	28.1	27.2	27.2	27.3	27.0	25.5	23.7
		MCDB	27.0	28.6	30.8	32.7	33.6	32.8	31.1	31.0	31.0	30.4	28.5	27.0
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.6	8.6	8.0	8.5	7.8	6.6	6.2	6.1	6.0	6.8	7.8
		MCDBR	8.7	10.0	9.4	8.7	12.3	10.3	8.5	7.7	7.6	7.5	7.7	8.4
		MCWBR	3.2	3.5	3.0	2.4	3.6	3.1	2.7	2.5	2.5	2.7	2.8	3.1
	5% WB	MCDBR	7.8	8.6	8.4	8.0	8.6	8.4	7.3	6.8	6.5	6.3	6.1	7.1
		MCWBR	3.2	3.5	3.1	2.6	3.1	3.2	2.9	2.8	2.7	2.7	2.7	3.2
Clear-Sky Solar Irradiance	taub	0.522	0.509	0.563	0.578	0.651	0.620	0.573	0.559	0.553	0.563	0.544	0.564	
	taud	1.975	2.030	1.923	1.928	1.779	1.859	1.989	2.028	2.041	2.000	1.987	1.900	
	Ebn at Noon	765	802	771	758	695	709	746	763	768	747	741	713	
	Edh at Noon	175	172	197	197	225	205	181	176	173	176	171	184	
All-Sky Solar Radiation	RadAvg	4.82	5.64	6.16	6.59	6.13	4.90	4.48	4.61	4.74	4.66	4.52	4.38	
	RadStd	0.17	0.21	0.23	0.24	0.22	0.41	0.53	0.37	0.39	0.43	0.33	0.28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.30	+0.32	N/A	N/A	+0.31	+0.30	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.30	+0.32	+0.48	N/A	+0.20	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page